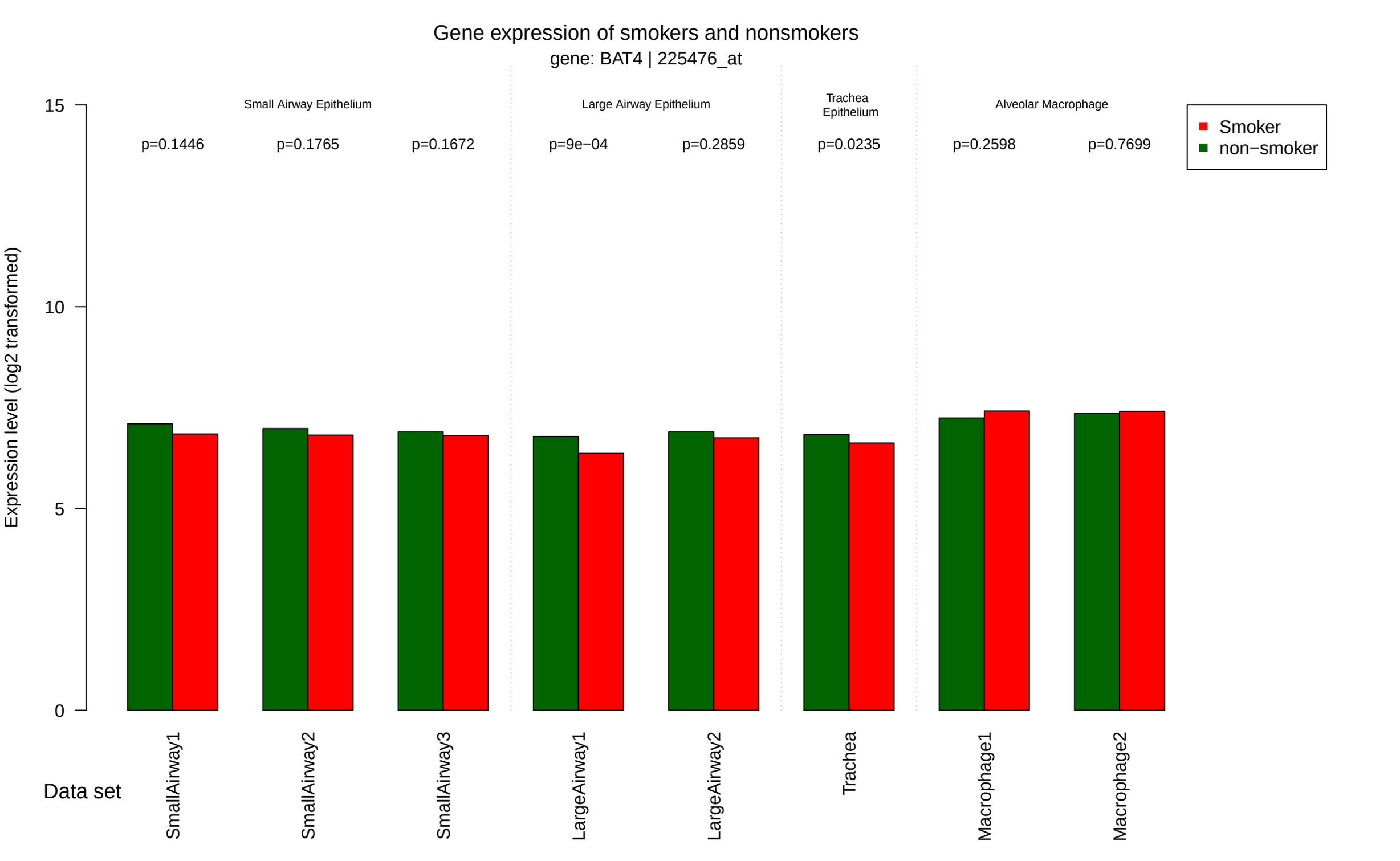
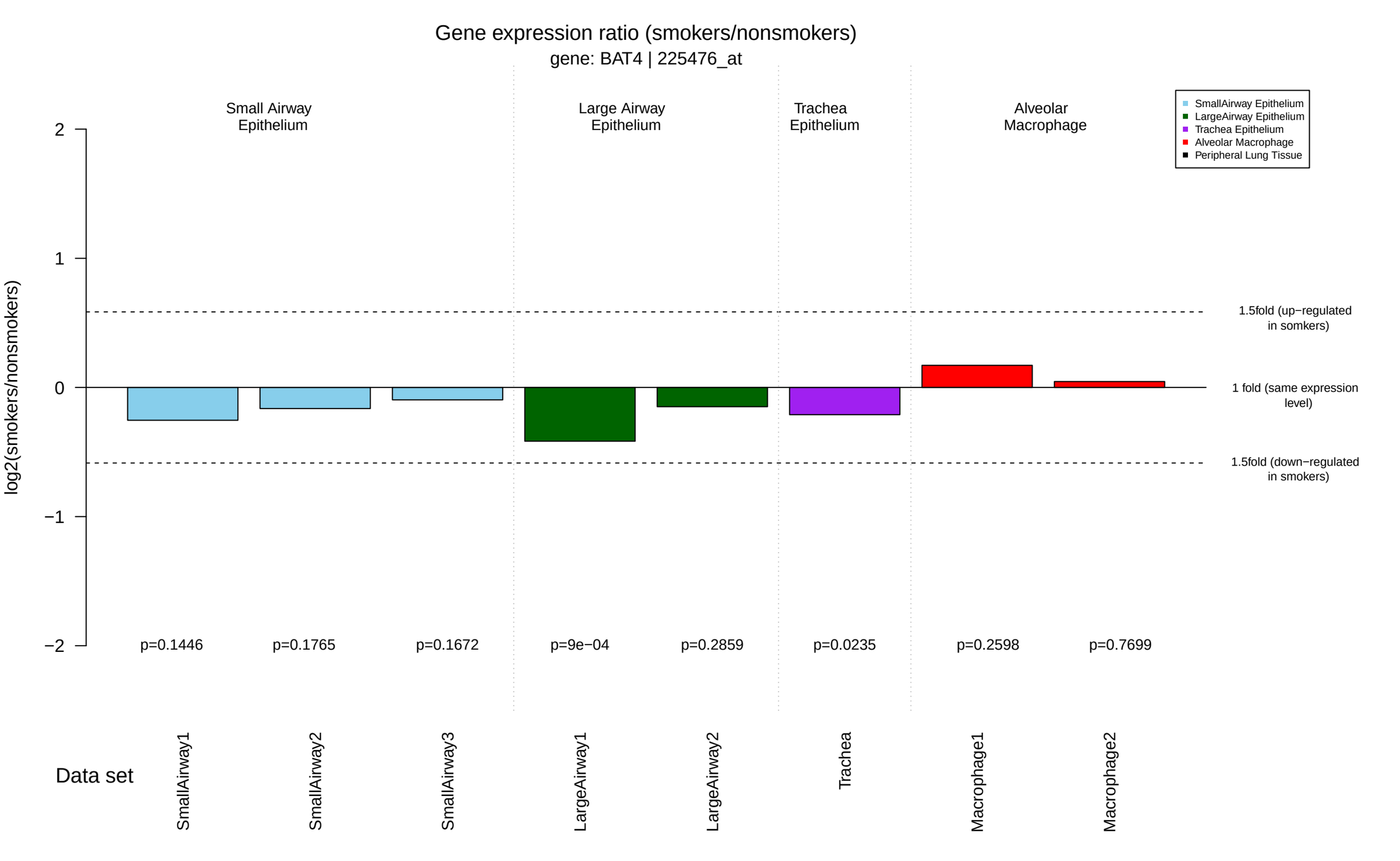
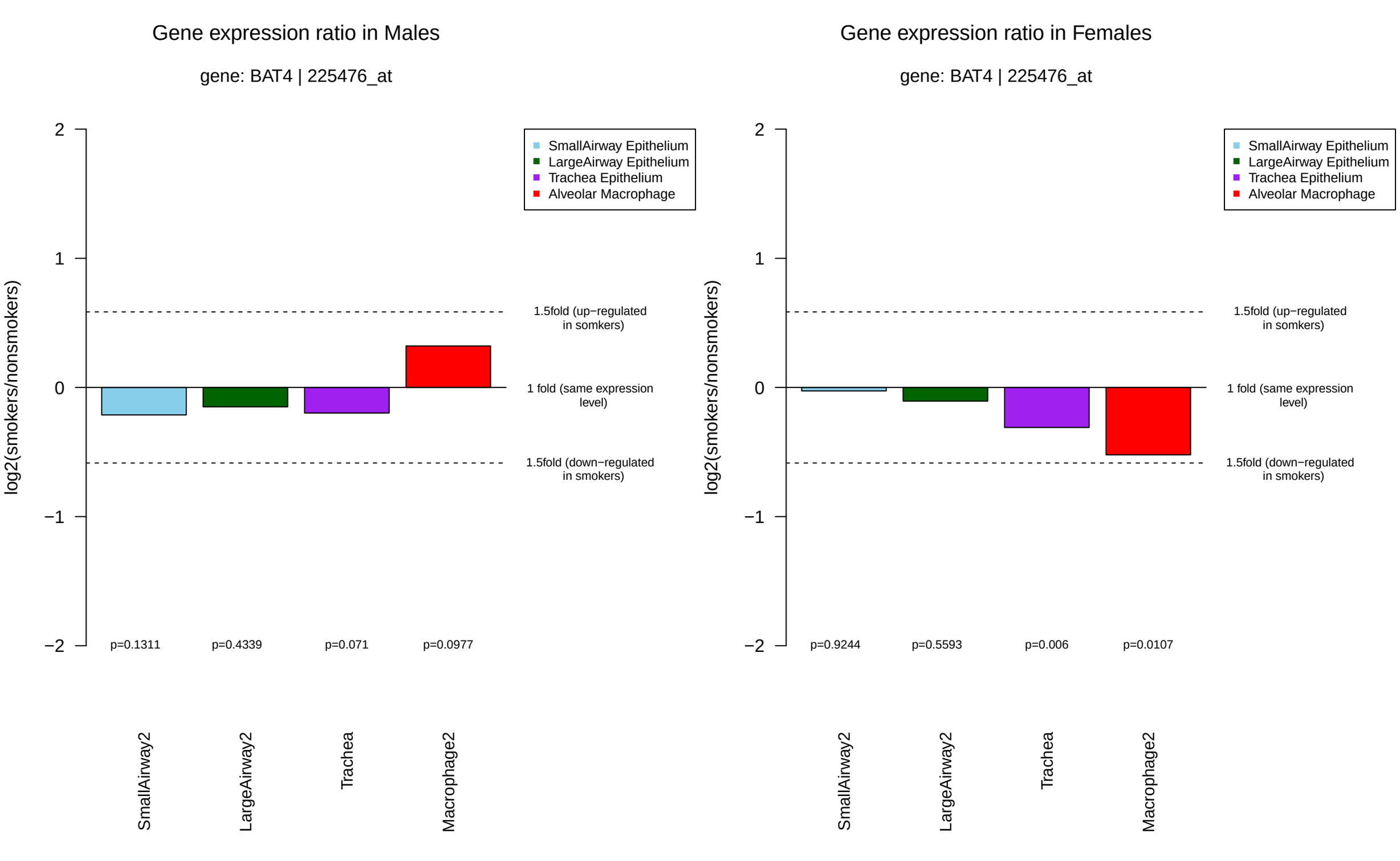
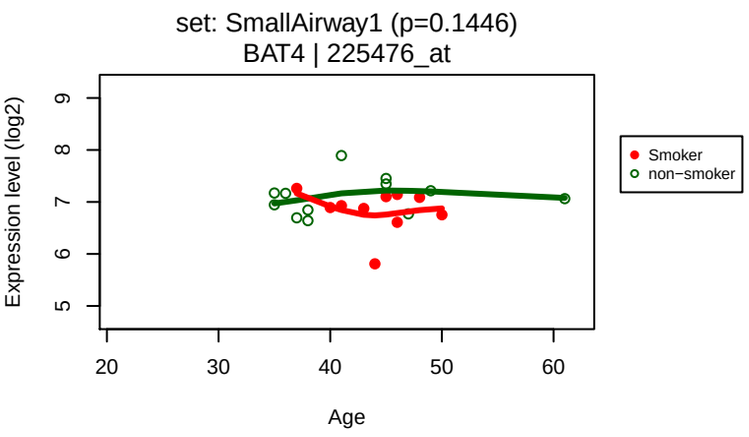
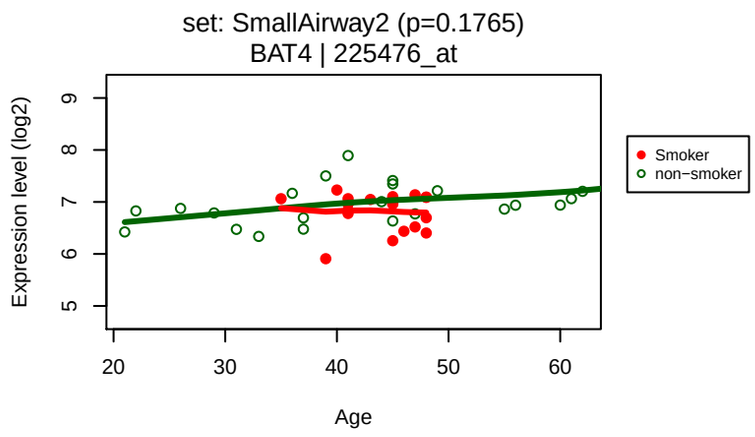
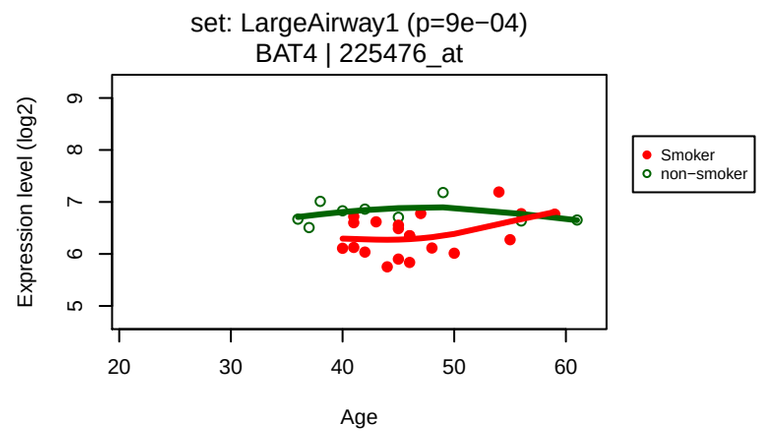

Gene expression of smokers and nonsmokers
gene: BAT4 | 225476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1446
p=0.1765
p=0.1672
p=9e−04
p=0.2859
p=0.0235
p=0.2598
p=0.7699
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

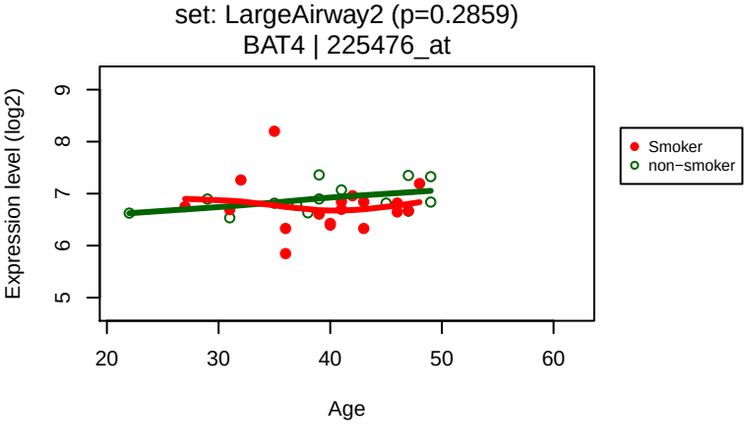

Gene expression ratio (smokers/nonsmokers)
gene: BAT4 | 225476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1446
p=0.1765
p=0.1672
p=9e−04
p=0.2859
p=0.0235
p=0.2598
p=0.7699
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

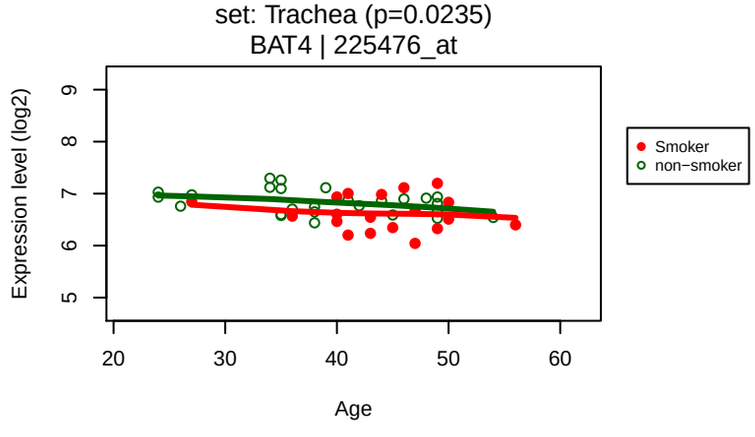

Gene expression ratio in Males
gene: BAT4 | 225476_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1311
p=0.4339
p=0.071
p=0.0977
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BAT4 | 225476_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9244
p=0.5593
p=0.006
p=0.0107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

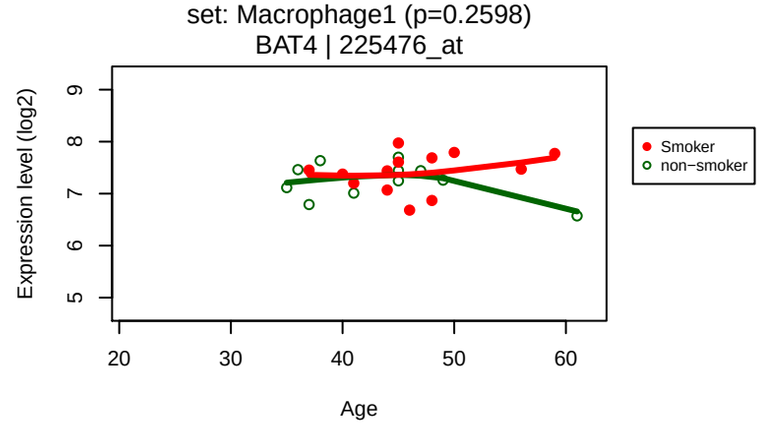

set: SmallAirway1 (p=0.1446)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker

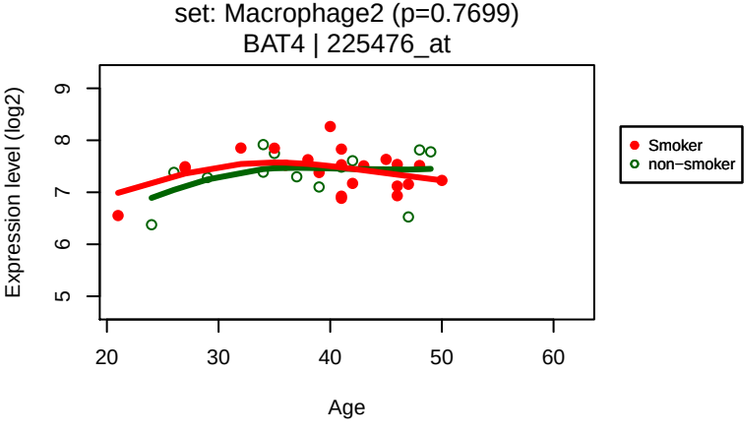

set: SmallAirway2 (p=0.1765)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker

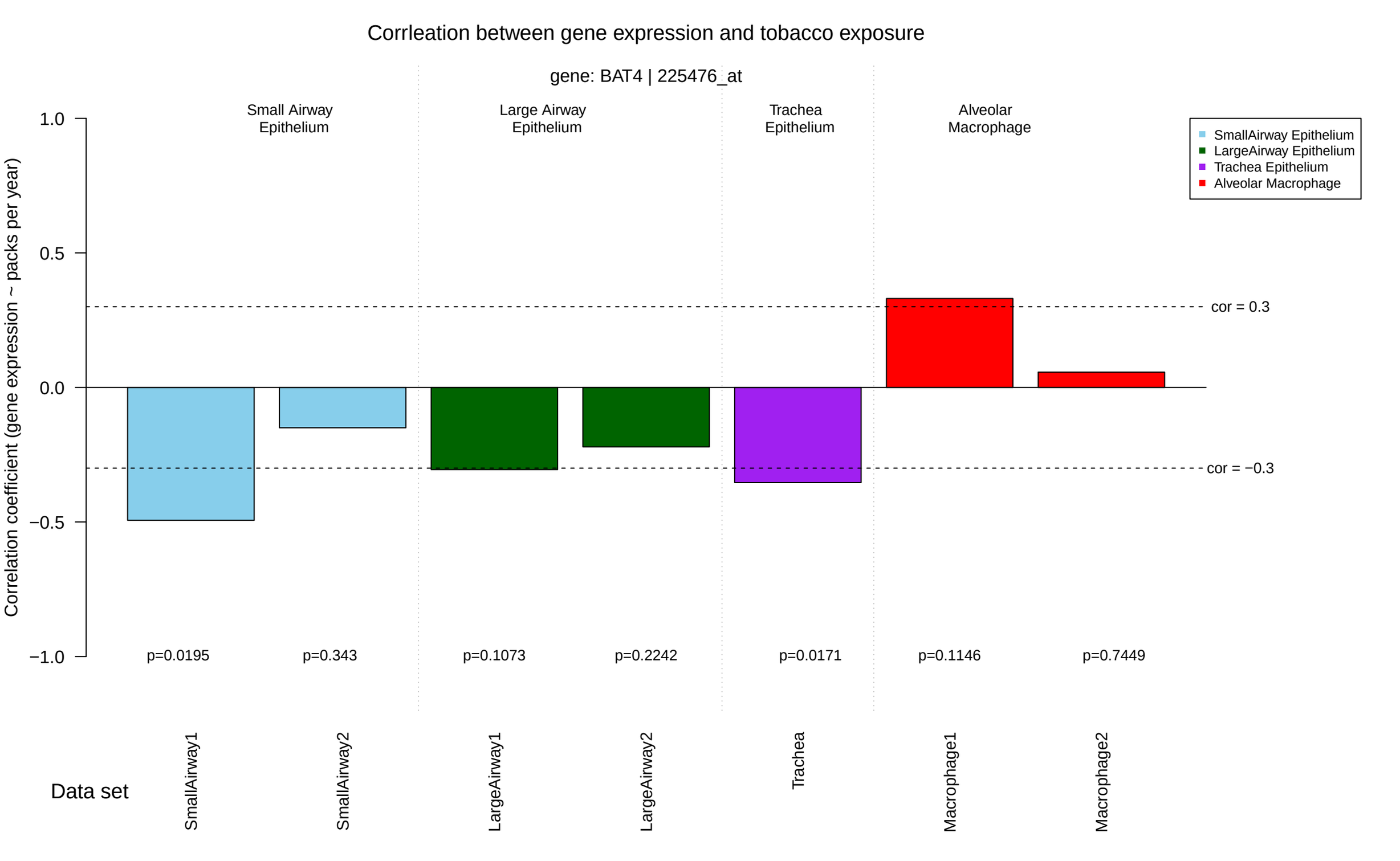

set: LargeAirway1 (p=9e−04)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2859)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0235)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2598)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7699)
BAT4 | 225476_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BAT4 | 225476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0195
p=0.343
p=0.1073
p=0.2242
p=0.0171
p=0.1146
p=0.7449
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2